

ELEC5305 PROJECT PROPOSAL — VERSION 2

Why Is Wav2Vec2 Robust to Lossy Audio Compression?

A Signal and Representation-Level Study of MP3 and Opus

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1. Project Overview

The original proposal asked when lossy compression begins to degrade automatic speech recognition (ASR). Preliminary MP3 and Opus experiments showed that this threshold-only framing was incomplete: measurable signal changes could appear while Word Error Rate (WER) remained close to the uncompressed WAV baseline. Following teaching-staff feedback, the project was therefore reframed to explain robustness rather than only locate a bitrate threshold.

Revised research question: How do MP3 and Opus compression alter the acoustic signal and the internal representations of Wav2Vec2, and which codec-induced distortions are associated with the onset of ASR errors?

The final analysis follows a three-level structure: signal distortion → representation drift → recognition degradation. The secondary question asks whether Wav2Vec2 remains robust because lossy codecs preserve information important to the recogniser even when measurable waveform or spectral distortion is already substantial.

2. Background and Motivation

Wav2Vec2 learns hierarchical speech representations directly from raw audio, making it possible to study not only the final recognition result but also how codec-induced perturbations propagate through internal layers. MP3 is used as a conventional lossy-audio reference, while Opus is included because it is designed for efficient speech and general audio communication at lower bitrates. The Opus range was extended below 16 kbps to include robust, transition and clearly degraded operating regions.

The project is grounded in Wav2Vec2 representation learning, perceptual audio coding, ASR robustness/domain shift and neural audio compression. Perceptual audio quality is treated as secondary: the primary objective is machine-recognition robustness.

3. Proposed Methodology and Feedback-Driven Refinements

A fixed pretrained WAV2VEC2_ASR_BASE_960H model is used without retraining or fine-tuning. LibriSpeech test-clean and test-other are evaluated with 500 fixed utterances per subset using random seed 5305. The same utterances are reused across WAV, MP3 and Opus conditions, enabling paired comparison.

- MP3: 128, 64, 32, 24 and 16 kbps. Opus: 64, 32, 16, 12, 8 and 6 kbps.
- Actual effective bitrate is measured as $8 \times \text{encoded file bytes} / \text{duration} / 1000$ rather than assuming the requested bitrate is achieved exactly.

- FFmpeg configuration is documented explicitly: libmp3lame for MP3 and libopus for Opus; comparisons are based on encoder settings, not file extensions alone.
- WER and word-level substitution/deletion/insertion counts are computed with JiWER. Paired bootstrap resampling provides 95% confidence intervals for Δ WER.
- Signal analysis uses log-spectral distortion, frequency-dependent distortion $D(f)$, effective retained bandwidth and delay-aware alignment.
- Representation analysis compares Wav2Vec2 Layers 1, 6 and 12 using cosine similarity and drift $= 1 - \text{cosine similarity}$.
- Representative individual failure cases are retained to complement aggregate statistics.

The working hypothesis is that compression may create measurable acoustic distortion before substantially changing deeper Wav2Vec2 representations or WER; under severe compression, late-layer drift may increase together with recognition errors. This is treated as an association to be tested, not as a causal assumption.

As an optional extension, EnCodec is evaluated at 24, 6 and 1.5 kbps. Because the mono EnCodec model operates at 24 kHz, an uncompressed $16 \rightarrow 24 \rightarrow 16$ kHz resampling control is included so that resampling effects are not confused with neural-codec effects.

4. Preliminary Results and Achievements to Date

The full MP3/Opus pipeline, paired statistical analysis, signal-level analysis, representation analysis, integrated analysis and failure-case analysis are operational and documented on the GitHub Project Site.

Dataset	Condition	WER	Δ WER	95% CI
test-clean	WAV	3.17%	0.00 pp	—
test-clean	Opus 6k	5.45%	+2.28 pp	[+1.86,+2.76]
test-other	WAV	8.26%	0.00 pp	—
test-other	Opus 6k	19.79%	+11.53 pp	[+10.30,+12.83]

Moderate compression often leaves WER near baseline even when measurable spectral distortion is present. Representation drift is generally largest in early layers and smaller in deeper layers under moderate compression, while severe compression increases late-layer drift together with WER. A particularly useful result is that Opus 6 and 8 kbps can show similar overall spectral-distortion magnitude but different representation drift and WER, indicating that a single signal-level metric does not fully explain ASR failure.

The optional EnCodec extension shows the same qualitative pattern: the resampling control has negligible effect, EnCodec 24 kbps remains close to baseline, and very low bitrate compression produces much larger representation drift and WER. This extension is treated as supporting evidence rather than a separate codec-ranking study.

5. Integration of Teaching-Staff Feedback

The feedback has been incorporated across the project rather than as isolated additions. The main changes include: revised title and research question; broader Opus bitrate coverage; measured actual bitrate; fixed-model and paired evaluation; a larger evaluation set and test-other condition; bootstrap uncertainty; S/D/I error analysis; quantitative spectral distortion, $D(f)$ and bandwidth

measures; delay/alignment checks; hidden-layer analysis at Layers 1/6/12; early-versus-late representation comparison; a signal → representation → task framework; representative failure cases; explicit FFmpeg reproducibility; clearer repository structure and execution order; literature grounding; a defined minimum core scope; and an optional EnCodec extension with a resampling control.

6. Expected Outcomes

The final deliverable will be a reproducible explanation of Wav2Vec2 robustness to lossy compression, supported by code, paired statistics, signal-level measurements, hidden-representation analysis, integrated figures and representative failure cases. The project contribution is therefore not simply a codec ranking, but an analysis of when signal degradation is absorbed by the recogniser and when it propagates into deeper representations and recognition errors.

7. Timeline and Current Status

Weeks	Task	Status
1–2	Topic selection and initial research question	Completed
3–5	Literature, dataset setup and baseline ASR	Completed
6–8	MP3/Opus implementation and bitrate experiments	Completed
8–10	test-other, bootstrap, error and signal analysis	Completed
9–11	Representation, integrated and failure-case analysis	Completed
10–11	Optional EnCodec extension and resampling control	Completed
11–12	Literature consolidation, interpretation and figure selection	In progress
12–13	Final report, GitHub documentation and presentation	Remaining

8. References

- [1] A. Baevski, Y. Zhou, A. Mohamed, and M. Auli, “wav2vec 2.0: A Framework for Self-Supervised Learning of Speech Representations,” *NeurIPS*, 2020.
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- [3] V. Panayotov, G. Chen, D. Povey, and S. Khudanpur, “LibriSpeech: An ASR Corpus Based on Public Domain Audio Books,” *ICASSP*, 2015.
- [4] J.-M. Valin, K. Vos, and T. Terriberry, “Definition of the Opus Audio Codec,” *RFC 6716*, 2012.
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